Thermal Equation of State and Critical Point

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***Abstract*—This report shows the derivation of th e Van der Waals constatns theoretically.**

I. Introduction

Provide physical background and motivation for the experiment. State the aim and what will be measured.

II. Theory

The Van der Waals equation of state for a real gas is

$$P = \frac{RT}{V-b} - \frac{a}{V^2}, \tag{1}$$

where P is pressure, T temperature, V molar volume, R the gas constant, and a, b are constants that account for intermolecular attraction and finite molecular size respectively.

At the critical point (P_c, T_c, V_c) the isotherm has a horizontal inflection, so

$$\left(\frac{\partial P}{\partial V}\right)_{T_c} = 0, \qquad \left(\frac{\partial^2 P}{\partial V^2}\right)_{T_c} = 0. \tag{2}$$

Compute the derivatives from the Van der Waals form:

$$\frac{\partial P}{\partial V} = -\frac{RT}{(V-b)^2} + \frac{2a}{V^3}, \tag{3}$$

$$\frac{\partial^2 P}{\partial V^2} = \frac{2RT}{(V-b)^3} - \frac{6a}{V^4}. \tag{4}$$

Applying the critical conditions at $V = V_c, T = T_c$ gives:

$$-\frac{RT_c}{(V_c-b)^2} + \frac{2a}{V_c^3} = 0, \tag{5}$$

$$\frac{2RT_c}{(V_c-b)^3} - \frac{6a}{V_c^4} = 0. \tag{6}$$

Dividing Eq. (6) by Eq. (5) yields

$$-\frac{2}{V_c-b} = \frac{3}{V_c}, \tag{7}$$

which gives

$$V_c = 3b. \tag{8}$$

Substituting $V_c = 3b$ into Eq. (5) leads to

$$T_c = \frac{8a}{27bR}. \tag{9}$$

Evaluating the Van der Waals equation at the critical point produces

$$P_c = \frac{a}{27b^2}. \tag{10}$$

Thus the standard critical-point relations are

$$V_c = 3b, \qquad P_c = \frac{a}{27b^2}, \qquad T_c = \frac{8a}{27bR}. \tag{11}$$

Solving for a and b from measurable critical constants (P_c, T_c) gives

$$b = \frac{RT_c}{8P_c}, \qquad a = \frac{27R^2T_c^2}{64P_c}. \tag{12}$$

The critical compressibility factor is

$$Z_c = \frac{P_cV_c}{RT_c} = \frac{3}{8}. \tag{13}$$

These relations are used to determine Van der Waals constants from experimental critical data.